Statistics of Financial Markets

Chapter 2: Statistical Distributions in Capital Markets

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Why Study Stable Distributions?

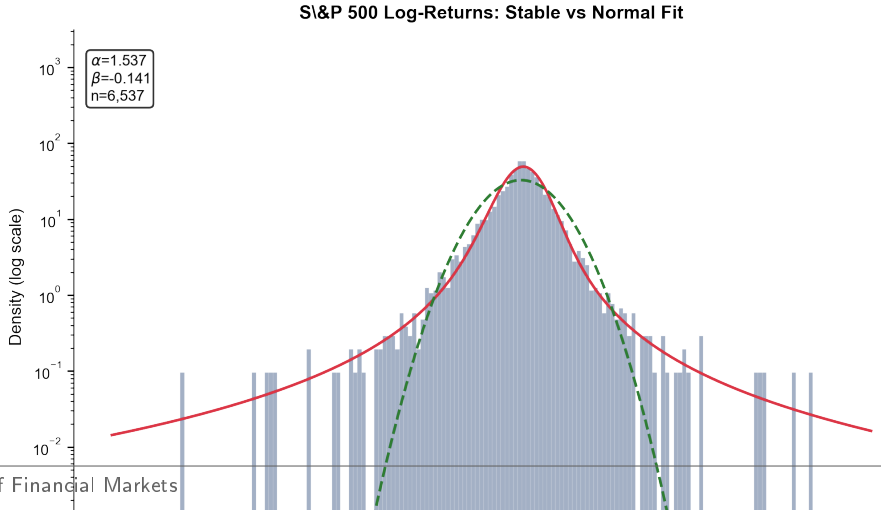
- Financial returns often exhibit **heavy tails** and **skewness** that the normal distribution cannot capture.
- Empirical evidence consistently shows that the Gaussian assumption underestimates the probability of extreme events (crashes, booms).
- **Stable distributions** provide a flexible framework for modeling such behavior.
- They generalize the normal distribution while allowing for:
  - Heavy tails (higher probability of extreme values)
  - Asymmetry (skewness in returns)
  - Varying degrees of peakedness
- Understanding these distributions is essential for accurate risk assessment in financial markets.

Why Study Stable Distributions?

- Stable distributions arise naturally from the **Generalized Central Limit Theorem** (GCLT).
- While the classical CLT leads to the normal distribution, the GCLT shows that sums of heavy-tailed random variables converge to stable laws.
- Key applications in finance:
  - **Risk management**: More accurate Value-at-Risk and Expected Shortfall estimates under heavy-tailed assumptions.
  - **Option pricing**: Models based on stable distributions capture the volatility smile observed in option markets.
  - **Portfolio theory**: Optimal allocation under non-Gaussian returns requires stable distribution frameworks.
- Mandelbrot (1963) first applied stable distributions to model cotton prices, showing that financial data deviate significantly from normality.
- Fama (1965) extended this analysis to stock market returns.

Motivating Example – Heavy Tails in Financial Returns

SFM_ch2_stable_fit



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Stable Distributions

- A family of probability distributions characterized by **four parameters**:
  - $\alpha \in (0, 2]$ – **stability index** (tail heaviness)
  - $\beta \in [-1, 1]$ – **skewness parameter**
  - $\gamma > 0$ – **scale parameter**
  - $\delta \in \mathbb{R}$ – **location parameter**

- Notation:

$$X \sim S(\alpha, \beta, \gamma, \delta)$$

- The normal distribution is a **special case** with $\alpha = 2$.
- For $\alpha < 2$, the distributions have **heavy tails** and **infinite variance**.
- Stable distributions are the only possible limiting distributions for properly normalized sums of i.i.d. random variables.

Historical Development

Year	Contributor	
1925	Paul Lévy	
1954	Gnedenko & Kolmogorov	
1963	Benoit Mandelbrot	Applied
1965	Eugene Fama	
1997–2003	John P. Nolan	

What Are Stable Distributions?

Definition 1 (Stable Distribution)

A random variable X is said to have a **stable distribution** if for any X_1, X_2 independent and identically distributed copies of X, and for any positive constants $a, b > 0$, there exist constants $c > 0$ and $d \in \mathbb{R}$ such that:

$$aX_1 + bX_2 \stackrel{d}{=} cX + d.$$

- The notation $\stackrel{d}{=}$ denotes **equality in distribution**.
- This property means that linear combinations of i.i.d. stable random variables have the **same type of distribution** (up to location and scale).
- The distribution is "stable" under addition – it preserves its form.
- This is a much stronger property than what holds for, e.g., the exponential or uniform distributions.

Defining Stability

- For a stable distribution, the constant c in the stability property satisfies:

$$c = (a^\alpha + b^\alpha)^{1/\alpha}$$

  where $\alpha \in (0, 2]$ is the **stability index** (also called the **characteristic exponent**).

- The value of α determines the tail behavior:
  - When $\alpha = 2$: **Gaussian distribution** – light tails, all moments exist.
  - When $\alpha < 2$: **Heavy-tailed distribution** – tails decay as a power law:

$$P(|X| > x) \sim C \cdot x^{-\alpha} \quad \text{as } x \to \infty.$$

- Moments: For $\alpha < 2$, only moments of order $p < \alpha$ are finite.
  - If $\alpha < 2$: variance is **infinite**.
  - If $\alpha \leq 1$: mean is **undefined**.

Generalized Central Limit Theorem (GCLT)

Theorem 1 (Generalized Central Limit Theorem)

If $X_1, X_2, \ldots$ are i.i.d. random variables with a distribution in the **domain of attraction** of a stable law, then there exist sequences of constants $a_n > 0$ and $b_n \in \mathbb{R}$ such that:

$$\frac{X_1 + X_2 + \cdots + X_n - b_n}{a_n} \xrightarrow{d} S(\alpha, \beta, \gamma, \delta).$$

- The classical **Central Limit Theorem** (CLT) is a special case with $\alpha = 2$:
  - When the X_i have finite variance σ^2, the normalized sum converges to $N(0, \sigma^2)$.
- The GCLT relaxes the finite variance assumption:
  - Heavy-tailed distributions (with infinite variance) can still have well-defined limits.
  - These limits are stable distributions with $\alpha < 2$.
- The normalizing constants satisfy $a_n \sim n^{1/\alpha}$ (rather than $a_n \sim \sqrt{n}$ for the classical CLT).

Parameterization Overview

- A stable random variable $X \sim S(\alpha, \beta, \gamma, \delta)$ is characterized by four parameters:

Parameter	Range	Role
α	$(0, 2]$	**Stability index** – controls tail heaviness
β	$[-1, 1]$	**Skewness parameter** – controls asymmetry
γ	$(0, \infty)$	**Scale parameter** – controls spread
δ	$(-\infty, \infty)$	**Location parameter** – shifts the distribution

- When $\beta = 0$, the distribution is **symmetric** about δ.
- When $\alpha = 2$, the parameter β becomes irrelevant (Gaussian is always symmetric).
- The parameterization is **not unique** – several conventions exist in the literature.

Effects of Parameters

Stability index $\alpha \in (0, 2]$

- Controls **tail heaviness**.
- Smaller $\alpha \Rightarrow$ heavier tails.
- $\alpha = 2$: Gaussian (lightest tails).
- $\alpha = 1$: Cauchy (very heavy tails).
- $\alpha < 1$: Extremely heavy tails.

Skewness $\beta \in [-1, 1]$

- Controls **asymmetry**.
- $\beta = 0$: symmetric distribution.
- $\beta > 0$: right-skewed (heavier right tail).
- $\beta < 0$: left-skewed (heavier left tail).
- $|\beta| = 1$: maximally skewed.

Scale $\gamma > 0$

- Controls the **spread** (width) of the distribution.
- Analogous to σ in the normal distribution.
- Larger $\gamma \Rightarrow$ wider distribution.

Location $\delta \in \mathbb{R}$

- **Shifts** the entire distribution.
- When $\alpha > 1$: δ is the mean.
- When $\alpha \leq 1$: δ is the median (if $\beta = 0$).

Probability Density Function

- In general, there is **no closed-form expression** for the probability density function (PDF) of a stable distribution.
- The PDF is obtained via the **inverse Fourier transform** of the characteristic function $\phi(t)$:

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \phi(t)\, e^{-itx}\, dt.$$

- Closed-form PDFs exist only for three special cases:
  - **Gaussian** ($\alpha = 2$): the normal distribution.
  - **Cauchy** ($\alpha = 1, \beta = 0$): the Cauchy distribution.
  - **Lévy** ($\alpha = 1/2, \beta = 1$): the Lévy distribution.
- For all other parameter values, the PDF must be computed **numerically**.
- Efficient algorithms for numerical computation were developed by Nolan (1997).

Special Case – Gaussian Distribution

- When $\alpha = 2$, the stable distribution reduces to the **normal (Gaussian) distribution**:

$$X \sim N(\delta,\ 2\gamma^2).$$

- **Characteristic function**:

$$\phi(t) = \exp\left(-\gamma^2 t^2 + i\delta t\right).$$

- **Probability density function**:

$$f(x) = \frac{1}{\sqrt{4\pi\gamma^2}} \exp\left(-\frac{(x-\delta)^2}{4\gamma^2}\right).$$

- Key properties:
  - **Symmetric** (β is irrelevant when $\alpha = 2$).
  - **Light-tailed**: exponential decay in the tails.
  - **All moments exist**: $\mathbb{E}[|X|^p] < \infty$ for all $p > 0$.
  - The **only stable distribution with finite variance**: $\mathrm{Var}(X) = 2\gamma^2$.

Normal Distribution – Simulation Example

SFM_ch2_stable_distributions

ch2_stable_distributions.png

Special Case – Cauchy Distribution

- When $\alpha = 1$ and $\beta = 0$, the stable distribution reduces to the **Cauchy distribution**:

$$X \sim \text{Cauchy}(\delta, \gamma).$$

- **Probability density function**:

$$f(x) = \frac{1}{\pi\gamma\left[1 + \left(\frac{x-\delta}{\gamma}\right)^2\right]}.$$

- Key properties:
  - **Symmetric** about δ and **heavy-tailed**: tails decay as x^{-2}.
  - **No mean**: $\mathbb{E}[|X|] = \infty$ (the integral diverges).
  - **No variance** or higher moments.
  - The **ratio of two independent standard normal** random variables follows a Cauchy distribution:

$$\text{If } Z_1, Z_2 \sim N(0,1) \text{ i.i.d., then } \frac{Z_1}{Z_2} \sim \text{Cauchy}(0,1).$$

  - The sample mean of Cauchy random variables has the **same distribution** as a single observation (no averaging effect).

Special Case – Lévy Distribution

- When $\alpha = 0.5$ and $\beta = 1$, we obtain the **Lévy distribution**:

$$X \sim S(0.5, 1, \gamma, \delta), \quad X > \delta$$

- **Probability density function**:

$$f(x) = \sqrt{\frac{\gamma}{2\pi}} \cdot \frac{e^{-\gamma/(2(x-\delta))}}{(x-\delta)^{3/2}}, \quad x > \delta$$

- Applications:
  - First passage times of Brownian motion.
  - Extreme value theory.

Tail Behaviour of Stable Distributions

- For $0 < \alpha < 2$, stable distributions exhibit **power-law tails**:

$$P(X > x) \sim C_\alpha(1+\beta)\,x^{-\alpha}, \qquad P(X < -x) \sim C_\alpha(1-\beta)\,x^{-\alpha}$$

- $C_\alpha = \dfrac{1-\alpha}{\Gamma(2-\alpha)\cos(\pi\alpha/2)}$.
- The smaller α, the heavier the tails.
- β introduces asymmetry: more mass on the left or right tail.
- Tail decay is **polynomial**, not exponential (unlike the Gaussian).
- Large-impact, low-probability events are more likely.

SFM_ch2_stable_fit

Moments and Their Existence

- Moment existence depends on α:

$$\mathbb{E}[|X|^p] < \infty \quad \text{for } 0 < p < \alpha; \qquad \mathbb{E}[|X|^p] = \infty \quad \text{for } p \geq \alpha$$

- The **mean** exists only if $\alpha > 1$.
- The **variance** exists only if $\alpha = 2$ (the Gaussian case).
- For Cauchy ($\alpha = 1$): no mean, no variance.
- For Lévy ($\alpha = 0.5$): even $\mathbb{E}[X]$ does not exist.
- This property makes stable distributions suitable for modelling extremes, but poses challenges for classical inference.

Stable PDFs – Visual Comparison

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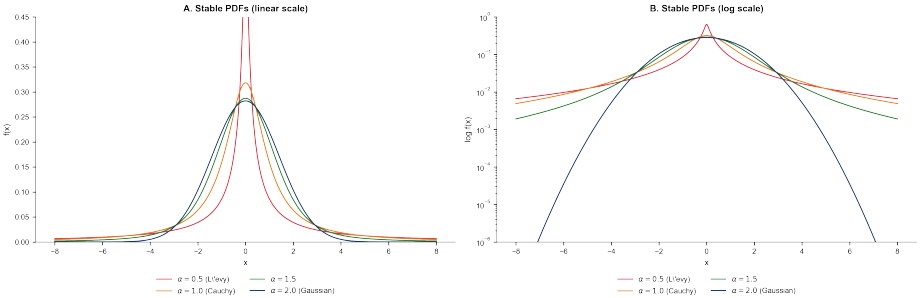

Left: linear scale. Right: log scale reveals tail differences. Financial returns typically have $\alpha \in [1.5, 1.9]$.

Application: S&P 500 Stable vs Normal Fit

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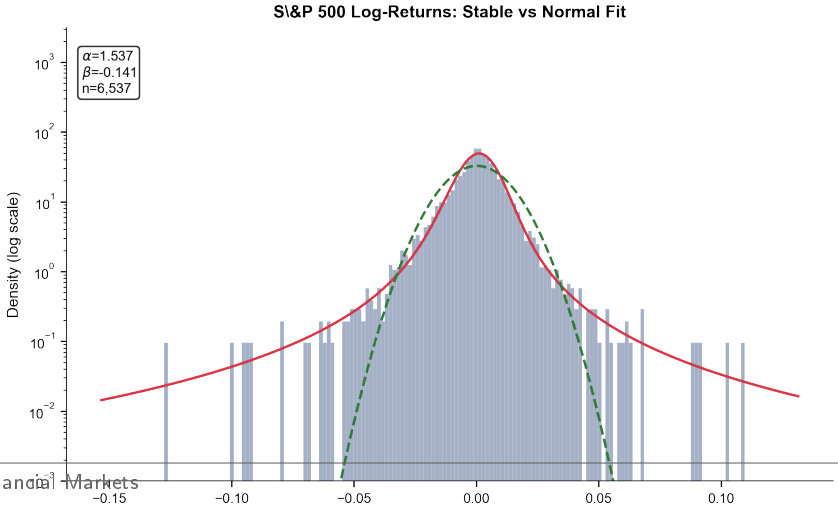

VaR: Stable vs Gaussian vs Empirical

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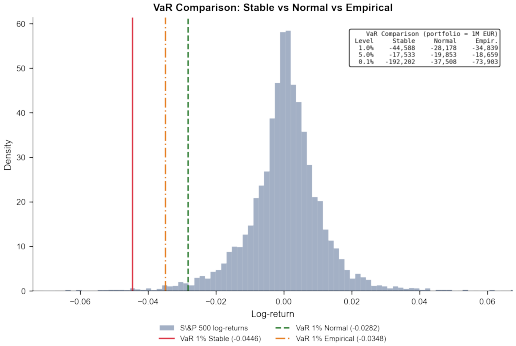

S&P 500 VaR Comparison

Level	Gauss	Stable	Emp.
5%	−1.9%	−2.1%	−2.1%
1%	−2.7%	−4.1%	−3.6%
0.1%	−3.6%	−9.0%	−6.9%

At 0.1%, stable VaR is ∼2.5× larger than Gaussian VaR.

Characteristic Function – Parameterization A

- The characteristic function of $X \sim S(\alpha, \beta, \gamma, \delta)$ under **Parameterization A** (Samorodnitsky and Taqqu, 1994):

Case $\alpha \neq 1$

$$\phi(t) = \exp\left\{-\gamma^\alpha |t|^\alpha \left(1 - i\beta \operatorname{sign}(t) \tan \frac{\pi\alpha}{2}\right) + i\delta t\right\}$$

Case $\alpha = 1$

$$\phi(t) = \exp\left\{-\gamma |t| \left(1 + i\beta \frac{2}{\pi} \operatorname{sign}(t) \ln |t|\right) + i\delta t\right\}$$

- Here $\operatorname{sign}(t) = \begin{cases} 1 & t > 0 \\ 0 & t = 0 \\ -1 & t < 0 \end{cases}$.

Characteristic Function – Parameterization B

- **Parameterization B** (Nolan, 1997) provides a form that is continuous in **all** parameters, including at $\alpha = 1$:

Case $\alpha \neq 1$

$$\phi(t) = \exp\left\{-\gamma^\alpha |t|^\alpha \left(1 + i\beta \operatorname{sign}(t) \tan\frac{\pi\alpha}{2}\left((\gamma|t|)^{1-\alpha} - 1\right)\right) + i\delta_B t\right\}$$

- This parameterization has the advantage that the characteristic function is **jointly continuous** in all four parameters $(\alpha, \beta, \gamma, \delta_B)$.
- It is particularly useful for **numerical computation** and **statistical estimation**.
- The location parameter δ_B differs from δ in Parameterization A.

Mapping Between Parameterizations

- The relationship between the location parameters in Parameterization A (δ_A) and Parameterization B (δ_B) is given by:

Conversion Formula

For $\alpha \neq 1$:

$$\delta_B = \delta_A + \beta\gamma \tan \frac{\pi\alpha}{2}$$

For $\alpha = 1$:

$$\delta_B = \delta_A + \beta\gamma \frac{2}{\pi} \ln \gamma$$

- When $\beta = 0$ (symmetric case): $\delta_A = \delta_B$ for all α.
- When $\alpha = 2$ (Gaussian case): $\delta_A = \delta_B$ regardless of β.
- Care must be taken when comparing results from different software packages, as they may use different parameterizations.

Estimation Methods

- Since no closed-form PDF exists in general, parameter estimation for stable distributions requires specialized methods.

Quantile-Based Methods (McCulloch, 1986)

- Use sample quantiles to estimate α and β.
- Fast and simple; good initial estimates.
- Based on relationships between quantile ratios and stable parameters.

Maximum Likelihood Estimation (Nolan, 2001)

- Numerically compute the PDF via Fourier inversion, then maximize the likelihood.
- Asymptotically efficient but computationally intensive.
- Requires accurate numerical integration routines.

Characteristic Function Methods (Koutrouvelis, 1980)

Simulation of Stable Random Variables

- The Chambers–Mallows–Stuck (CMS) algorithm (1976) generates standard stable random variables $Z \sim S(\alpha, \beta, 1, 0)$:

CMS Algorithm

1. Generate $U \sim \text{Uniform}\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ and $W \sim \text{Exponential}(1)$, independently.
2. For $\alpha \neq 1$, compute:

$$Z = S_{\alpha,\beta} \cdot \frac{\sin(\alpha(U + B_{\alpha,\beta}))}{(\cos U)^{1/\alpha}} \cdot \left(\frac{\cos(U - \alpha(U + B_{\alpha,\beta}))}{W}\right)^{(1-\alpha)/\alpha}$$

   where $B_{\alpha,\beta} = \frac{1}{\alpha} \arctan\left(\beta \tan \frac{\pi\alpha}{2}\right)$ and $S_{\alpha,\beta} = \left(1 + \beta^2 \tan^2 \frac{\pi\alpha}{2}\right)^{1/(2\alpha)}$.
3. Scale and shift: $X = \gamma Z + \delta$.

- Available in Python via `scipy.stats.levy_stable` and in R via the `stabledist` package.

Summary

- Stable distributions provide a **theoretically grounded** and **flexible** framework for modeling financial returns that exhibit heavy tails and skewness.

- Key takeaways:
  - The **Generalized Central Limit Theorem** justifies the use of stable distributions as limits of sums of i.i.d. heavy-tailed variables.
  - Four parameters $(\alpha, \beta, \gamma, \delta)$ control tail behavior, asymmetry, scale, and location.
  - The **Gaussian** $(\alpha = 2)$ and **Cauchy** $(\alpha = 1, \beta = 0)$ distributions are important special cases.
  - **No closed-form PDF** in general – numerical methods are essential.
  - Multiple **parameterizations** exist; care is needed when comparing results.
  - Efficient **estimation** (MLE, quantile methods) and **simulation** (CMS algorithm) methods are available.

- Stable distributions remain an active area of research in **financial econometrics**, **risk management**, and **quantitative finance**.

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